

# PAPER\_1507 — F\_U Temporal Decay $\alpha = 1/\text{SO}_5^3 = 0.001/\text{day}$ EXACT

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational F\_U dynamics) **Date:** June 18, 2026 **Status:** CLOSED — Two-stage F\_U slow temporal decay rate

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## Observation

PAPER\_1080 (Two-Stage F\_U Refinement) introduces a two-tier temporal evolution of the F\_U master equation: - **Stage 1:**  $F_U^{(1)} = \sum_{k=1}^3 (U_{g,k} - U_{b,k}) + U_m + A_{\mu\nu}$  - **Stage 2:**  $U_{g1}(t) = U_{g1,0} \cdot e^{-\alpha \cdot t}$  with  $\alpha = 0.001 \text{ day}^{-1}$

The slow decay rate  $\alpha$  is distinct from the canonical  $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$  used in Holmlid/Parkhomov LENR (the two rates differ by a factor of exactly 2).

## UQFF Closed Identity

$$\alpha = 1 / \text{SO}_5^3 = 1 / 1000 = 0.001 \text{ day}^{-1} \quad \text{EXACT}$$

## Physical Interpretation

The F\_U temporal decay rate is exactly inverse-cubic in SO\_5 — one inverse power per spatial dimension of the SO(5) symmetry projected onto F\_U evolution. Equivalent forms:

- $\alpha \cdot \text{day} = 10^{-3}$  exactly
- 1000 days for U\_g1 to decay by factor e
- $\approx 2.74$  years for one e-folding

The doubling relationship  $\alpha = 2\kappa$  links F\_U gravitational decay (slow) to LENR Holmlid decay (canonical  $\kappa$ ), with the factor 2 corresponding to the two-fold structural pairing in DPM (CW + CCW).

## NOT REPLACEMENT

SM treats decay rates as fit constants per system. UQFF supplies a single integer-primitive structural value for the F\_U slow-decay channel.

## Reference

- Source: PAPER\_1080 TwoStage F\_U Refinement
- Related:  $\kappa = 5 \times 10^{-4} / \text{day}$  Holmlid canonical, PAPER\_1141 (Rossi variants)
- Calculator dispatch: `calculate_paradox({"paradox": "f_u_alpha_decay_1_over_so5_3"})`

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